

35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — KILOMETERS (1 kilometer = 100,000 centimeters)—continued								
FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION		TO FIND KILOMETERS PER CENTIMETER (CM) (x in ratio)	KILOMETERS PER CM	TOTAL KILOMETERS ON SCALE	TO FIND TOTAL SCALE LENGTH IN CENTIMETERS (y in ratio)	TOTAL SCALE LENGTH IN	
	Scale Unit	represents Map Unit	Use ratio below or $\frac{\text{SCALE}}{100\,000}$			Use ratio below or $\frac{\text{Kilometers on scale}}{\text{Kilometers per cm}}$	CENTI-METERS	MILLI-METERS
1:250 000	1 cm	250 000 cm	$\frac{100\,000}{1} = \frac{250\,000}{x}$	2.500	30	$\frac{2.500}{1} = \frac{30}{y}$	12.000	120.00
1:400 000	1 cm	400 000 cm	$\frac{100\,000}{1} = \frac{400\,000}{x}$	4.000	60	$\frac{4.000}{1} = \frac{60}{y}$	15.000	150.00
1:500 000	1 cm	500 000 cm	$\frac{100\,000}{1} = \frac{500\,000}{x}$	5.000	60	$\frac{5.000}{1} = \frac{60}{y}$	12.000	120.00
1:750 000	1 cm	750 000 cm	$\frac{100\,000}{1} = \frac{750\,000}{x}$	7.500	60	$\frac{7.500}{1} = \frac{60}{y}$	8.000	80.00
1:1 000 000	1 cm	1 000 000 cm	$\frac{100\,000}{1} = \frac{1\,000\,000}{x}$	10.000	125	$\frac{10.000}{1} = \frac{125}{y}$	12.500	125.00
1:2 000 000	1 cm	2 000 000 cm	$\frac{100\,000}{1} = \frac{2\,000\,000}{x}$	20.000	300	$\frac{20.000}{1} = \frac{300}{y}$	15.000	150.00
1:2 500 000	1 cm	2 500 000 cm	$\frac{100\,000}{1} = \frac{2\,500\,000}{x}$	25.000	400	$\frac{25.000}{1} = \frac{400}{y}$	16.000	160.00
1:5 000 000	1 cm	5 000 000 cm	$\frac{100\,000}{1} = \frac{5\,000\,000}{x}$	50.000	800	$\frac{50.000}{1} = \frac{800}{y}$	16.000	160.00
1:7 500 000	1 cm	7 500 000 cm	$\frac{100\,000}{1} = \frac{7\,500\,000}{x}$	75.000	800	$\frac{75.000}{1} = \frac{800}{y}$	10.666	106.66
1:10 000 000	1 cm	10 000 000 cm	$\frac{100\,000}{1} = \frac{10\,000\,000}{x}$	100.000	1500	$\frac{100.000}{1} = \frac{1500}{y}$	15.000	150.00
To find kilometers per centimeter on 1: 250 000 map . . . 100 000 centimeters = 1 kilometer Show in ratio as ... $\frac{100\,000}{1} \frac{\text{centimeters}}{\text{kilometers}}$ Let SCALE (250 000) be in centimeters Fractional scale says 1 centimeter represents 250,000 centimeters Let x be kilometers that 1 cm represents on map Show in ratio as ... $\frac{250\,000}{x} \frac{\text{centimeters}}{\text{kilometers}}$ Solution . . . $\frac{100\,000}{1} = \frac{250\,000}{x}$ $\frac{100\,000}{100\,000} x = \frac{250\,000}{100\,000}$ $x = \frac{250\,000}{100\,000} \text{ (SCALE)}$ $x = 2.5$ 100 000 • x = 250 000 • 1								